TA390

User Manual · Firmware v1.3 · Program v1.3

For the TYT MD-UV380 / MD-UV390 handheld radios

DMR / FM firmware based on OpenGD77, flashing and codeplug (CPS) programs

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October 2026 · Based on the OpenGD77 OpenMDUV380 source code (30.01.2026) · Not an official OpenGD77 release

1. Introduction

TA390 is a modified version of the open source **OpenGD77** firmware for the TYT MD-UV380 / MD-UV390 dual band DMR handheld radios. The main OpenGD77 features are kept: DMR (two timeslots), FM, channel and VFO modes, scanning, spectrum, APRS, GPS, satellite tracking and voice prompts. The radio menus can be Turkish or English (chosen when flashing).

On the computer side there is the **TA390 program** (Windows) and the **TA390 app** (Android): they flash the firmware, make and restore backups, upload a boot image and the satellite data, and include a **CPS** to edit the channels, zones, contacts, APRS and radio settings.

Main differences from OpenGD77:

- **Dual Watch:** two stored channels of the zone are shown one under the other and both are monitored (like the original TYT firmware).
- **Speaker and missed call icons:** show which channel the audio comes from and which channel had a call.
- **Logs:** received DMR and FM calls are stored with date, time, length and channel.
- **APRS:** the position is sent independently of the open channel. **APRS messages** can be sent from the radio.
- **Screen saver:** when the backlight goes off, clock, date and battery at low brightness.
- **Gps/Location menu:** GPS on / off with the Green key.
- **TG lists menu:** the TG lists are edited on the radio.
- **Shortcuts menu:** all the key shortcuts can be read on the radio screen.
- **Games:** Snake, Tetris and Mines (the radio keeps listening).
- **Simple header line:** mode, TS and colour code, power, APRS and GPS icons, battery.
- **Screen wake up:** while the screen is dark the first key press only turns the screen on.

Removed to keep it simple: roaming, channel distance, channel lock, key timing settings, theme menu, automatic night mode, inverted colours, minimum brightness, clock in the header and per channel APRS settings.

> **Warning:** This firmware is not an official OpenGD77 release; it is not made, endorsed or supported by the OpenGD77 developers. It is for amateur radio use. When transmitting, follow the regulations and the frequency / power limits of your licence. Always make a backup before flashing.

> **Note:** The screenshots in this manual show the Turkish interface. The programs (Settings › Program language) and the radio (radio language when flashing) can be used in English.

2. Package and files

File	Content
`TA390_Kurulum_v1.3.exe`	Setup of the TA390 program for Windows.
`TA390_v1.3.apk`	TA390 app for Android phones (CPS, firmware flashing / update and backups over a USB OTG cable).
`TA390_v1.3.zip`	Radio firmware `TA390_v1.3.bin`, language files (`*.gla`), `BENIOKU.txt` and the `LISANSLAR` (licences) folder.
Codeplug file	`.ta390` files are opened in the CPS. The `.g77` files of the OpenGD77 CPS can be opened too.

Donor file: the audio codec needed for DMR (AMBE) is not distributed with TA390. For the first installation the official **MD-9600 v26.45** firmware file (the donor, also used by the OpenGD77 CPS) is needed; the program takes the codec from it and stores the donor. If no donor is stored, the program downloads it automatically from the link given in the OpenGD77 installation guide. Do not give your donor file to others.

3. TA390 program: installation

- Run `TA390_Kurulum_v1.3.exe`. The program is installed in `Program Files\TA390`, with shortcuts in the Start menu and on the desktop. An old “TA390 Yükleyici” is removed.
- To uninstall, use the Windows Apps list or `Kaldir.exe` in the program folder.
- Settings, the stored donor and the automatic backups are kept in `%APPDATA%\TA390`.
- The program has four tabs: **CPS**, **Install / Update**, **Backup and more**, **Settings**.

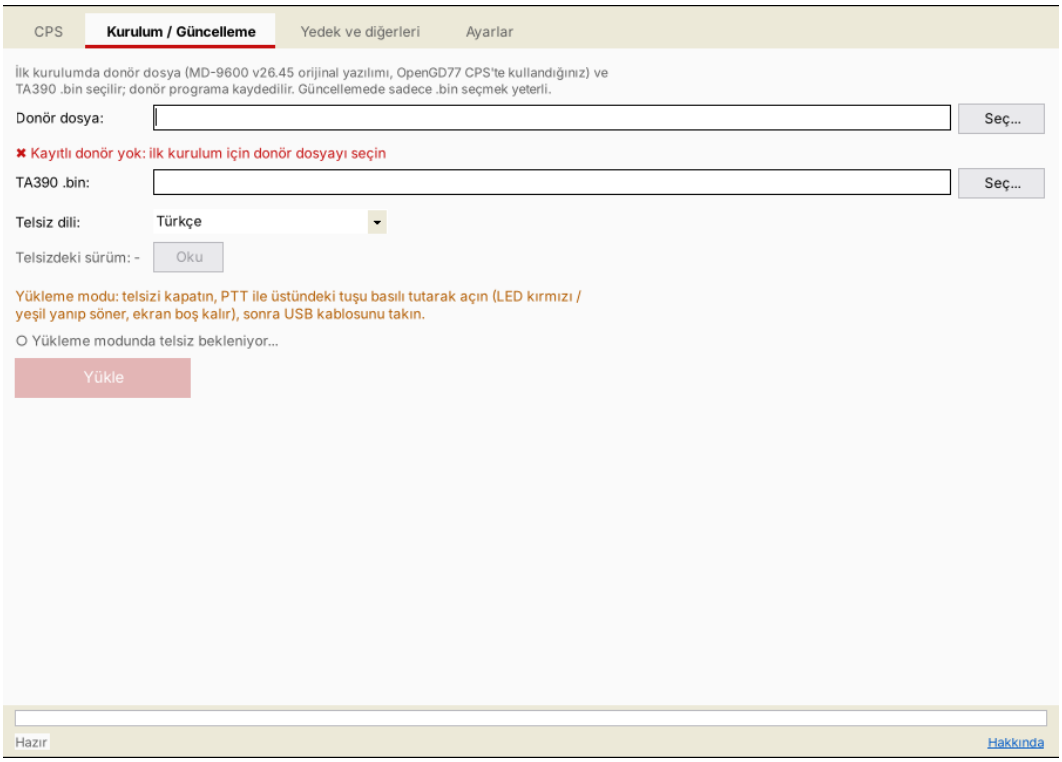
Drivers

- **Radio switched on normally** (CPS, backups, boot image, satellites): Windows sees it as a serial port (COM); no extra driver is needed.
- **Radio in bootloader mode** (flashing, MCU ROM restore): the STM DFU (DfuSe) driver installed by the OpenGD77 CPS is used. Without the OpenGD77 CPS, `STDFU.dll` and `STTubeDevice30.dll` can be copied into the TA390 folder. For radios with the WinUSB (Zadig) driver, `libusb-1.0.dll` comes with the program.

Android app

- Install `TA390_v1.3.apk` on the phone (Android may ask to allow installing from unknown sources). A phone with USB OTG support and an OTG cable / adapter are needed.
- The app has the same tabs and functions as the Windows program: **CPS**, **Install / Update**, **Backup and more**, **Settings**. Channels and contacts are edited in a window opened by tapping them; the order is changed by dragging the ≡ handle of a row.
- When the radio is plugged in, Android asks for the USB permission: allow it. The radio in bootloader mode asks for its own permission. No driver is needed.
- The donor is downloaded and stored by the app at the first flashing. The codeplug backup made before writing is saved in `Android/data/com.tb1yzc.md390cps/files/backups`.
- The buttons working with the radio (backups, boot image, satellites, clock) stay disabled until the radio is connected.

4. Flashing and updating the firmware



Install / Update tab (Turkish interface)

Bootloader mode

Switch the radio off. Switch it on while holding **PTT and the key just above it** (SK1): the LED flashes red / green and the screen stays blank. Then plug the USB cable. When the program finds the radio it shows “Radio in bootloader mode, ready”.

First installation (from the original firmware or from OpenGD77)

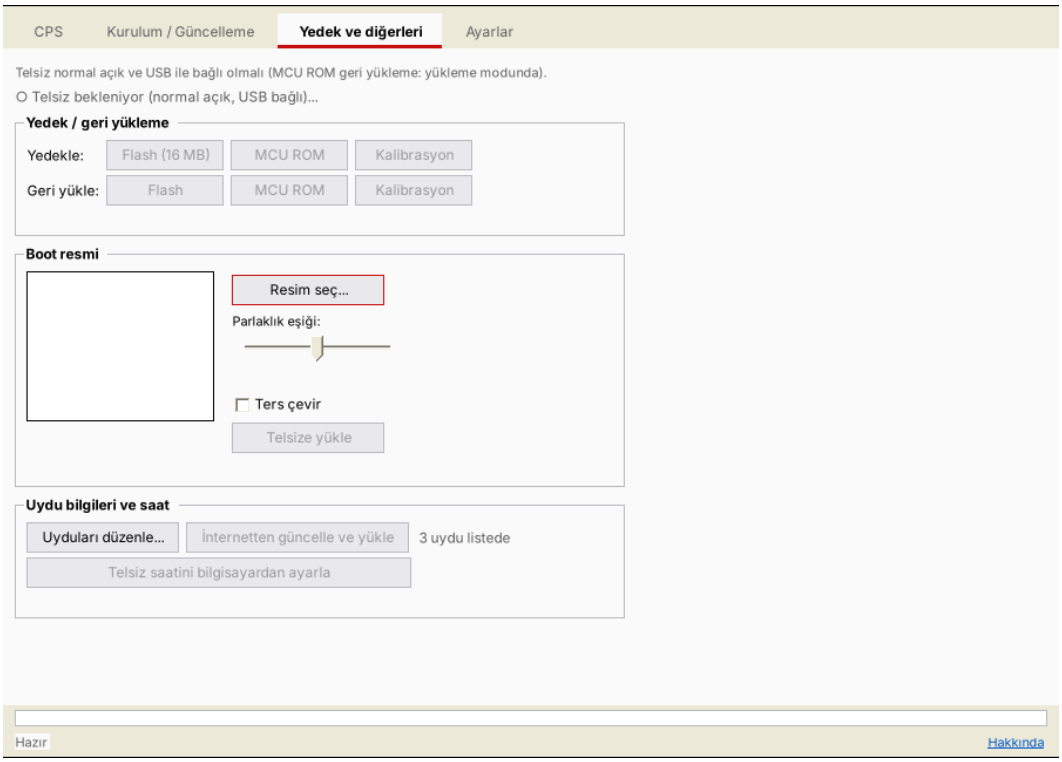
- **TA390 firmware**: select the `TA390_v1.3.zip` package directly, like with OpenGD77 (its .bin and language file are used). A `.bin` taken out of the zip can be selected too.
- **Donor file**: not needed if you let the program download it. You can also select the MD-9600 v26.45 donor yourself; a green tick shows a valid file.
- **Radio language**: Turkish or English. The chosen language is flashed with the firmware.
- Press **Flash** and confirm. Do not unplug the cable or switch the radio off while flashing. The radio restarts when done.

Update

Once the donor is stored, selecting the new `.zip` package and pressing **Flash** is enough. The **Read** button of the **Firmware in the radio** line (radio switched on normally) shows the installed version.

> **Note:** Without a donor the firmware works without DMR audio (FM only). The program asks before flashing.

5. Backup and more



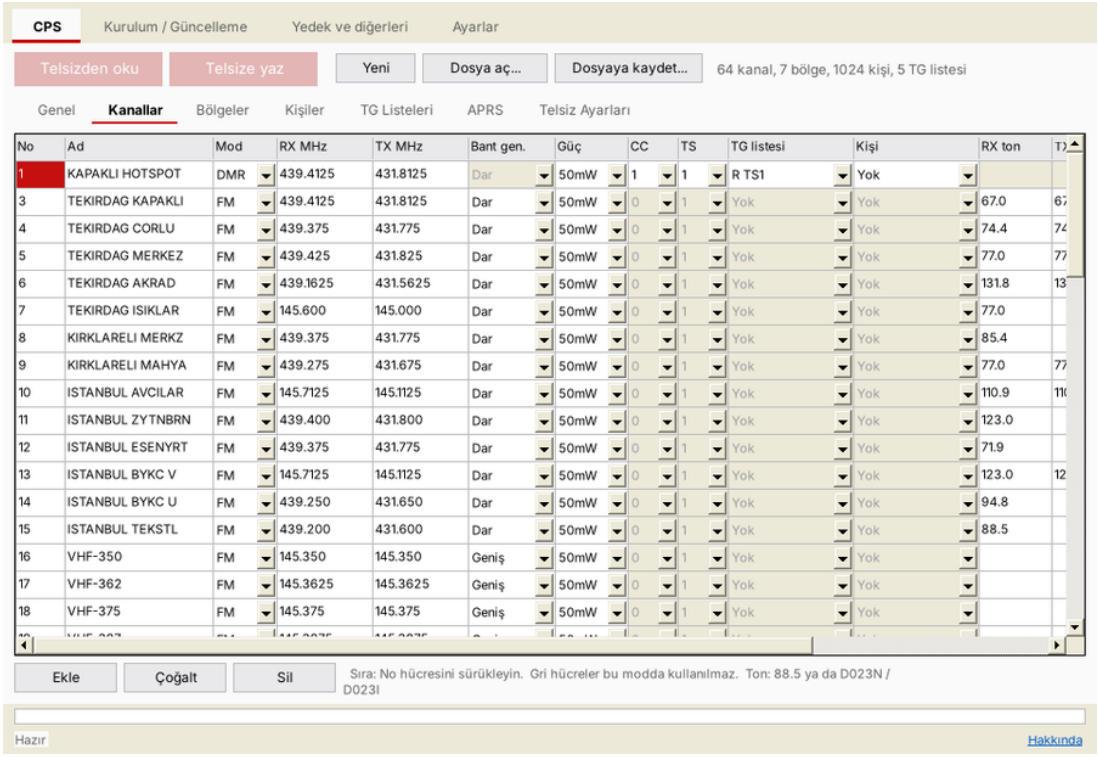
Backup and more tab (Turkish interface)

For this tab the radio must be **switched on normally and connected with USB** (except MCU ROM restore, which is done in bootloader mode).

Function	Description
Backup: Flash (16 MB)	The whole SPI flash of the radio: channels, settings, logs, DMR ID database, calibration. Takes a few minutes.
Backup: MCU ROM	Copy of the firmware in the processor. Keep it to go back to the original firmware.
Backup: Calibration	The factory calibration of the radio (1 KB). Each radio has its own.
Restore: Flash	Writes a flash backup back. Only the different parts are written. Everything in the radio is replaced by the backup.
Restore: MCU ROM	Flashes the firmware of an MCU ROM backup (in bootloader mode).
Restore: Calibration	Only use a backup of the same radio; the calibration of another radio gives wrong power and frequency.
Boot image	Select a picture (png, jpg, bmp, gif). It is converted to 128×64 black and white; adjust it with **Brightness threshold** and **Invert**. **Upload to radio** writes it. The radio shows it full screen at power on (CPS › General › Boot screen: Picture).
Satellites	Choose the list with **Edit satellites**; **Download and upload** (Android: **Update from internet**) downloads the current orbit data (TLE) and uploads it to the radio; the radio clock is set too.
Radio clock	Sets the radio clock from the computer / phone.

> **Important:** Before flashing for the first time, make the **Flash**, **MCU ROM** and **Calibration** backups and keep them in a safe place.

6. CPS: editing the codeplug



CPS › Channels (Turkish interface)

The CPS tab edits the codeplug of the radio (general data, channels, zones, contacts, TG lists, APRS and radio settings). Buttons at the top:

Button	Function
Read radio	Reads the codeplug of the radio (radio switched on, USB connected).
Write radio	Writes the codeplug to the radio (see below).
New	Creates an empty codeplug.
Open file...	Opens a `.ta390` file or a `.g77` file of the OpenGD77 CPS.
Save file...	Saves the codeplug to a `.ta390` file.

- You can work **without the radio**: start with **New** or **Open file**, edit, save to a file, write it to the radio later.
- With unsaved changes, closing the program, opening another file or reading the radio asks “Save to a file?”.
- An OpenGD77 CPS file (`.g77`) is saved as `.ta390`.

Writing to the radio

- The program first reads the codeplug of the radio and makes an **automatic backup** in `%APPDATA%\TA390\backups`.
- Then only the parts edited in the CPS (channels, zones, contacts, TG lists, APRS, callsign, DMR ID, boot texts) are put onto the content of the radio and **only the changed sectors** are written. The other data of the radio (VFO, quick keys, band limits...) are kept.
- The radio restarts when done. If radio settings were changed too, it restarts once more (see below).

General

First tab: callsign, DMR ID, boot screen type (Text / Picture) and the two boot text lines.

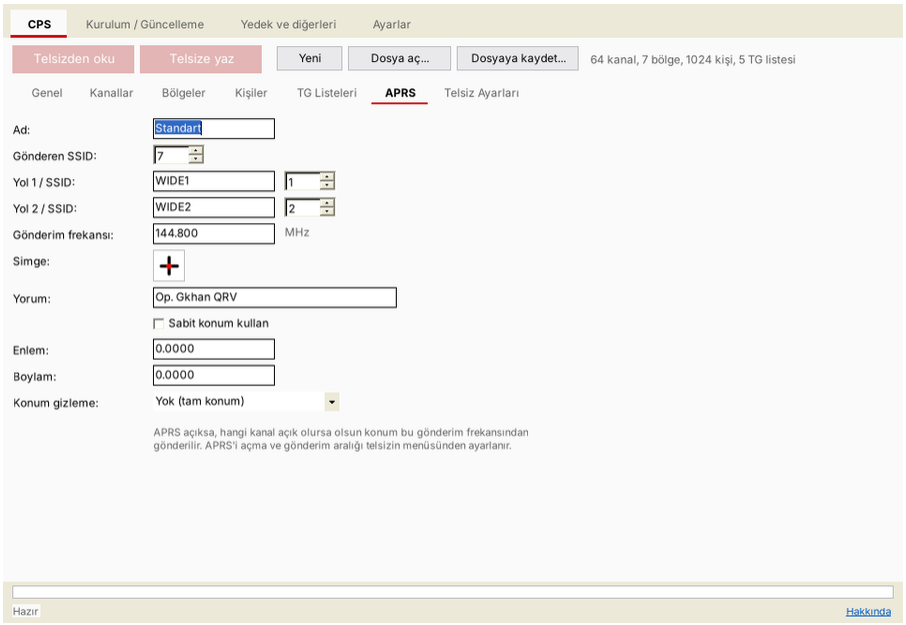
Channels

- The cells are edited directly. **Add**, **Duplicate** and **Delete** are at the bottom. Hovering over a column title shows its description.
- **Grey cells** are not used in the mode of the channel: CC (colour code), TS, TG list and Contact on FM channels; bandwidth and tones on DMR channels. The contact is used on DMR channels without a TG list. Squelch is not set per channel (general setting of the radio).
- **Rx only** and **zone / all scan skip** are not in the CPS; they are set in the channel details of the radio (the codeplug values are kept).
- Tones: `88.5` for CTCSS, `D023N` / `D023I` for DCS; empty means no tone.
- **Order:** drag the **No** cell at the left of the row; a red line shows where it goes. The zones keep the same channels.

Zones, Contacts, TG Lists

- **Zones:** select a zone in the left list; the middle list has the channels of the zone, the right one all the channels. Channels are added with ◄ **Add** / **Remove** ► or by drag and drop. The order of the channels is changed with ▲▼ or by dragging, the order of the zones by dragging in the left list.
- **Contacts:** name, ID, type (Group, Private, All call) and TS override. Drag the No cell to change the order; channels and TG lists follow.
- **TG Lists:** 32 TGs per list maximum. Adding, removing and ordering work like the zones.

APRS

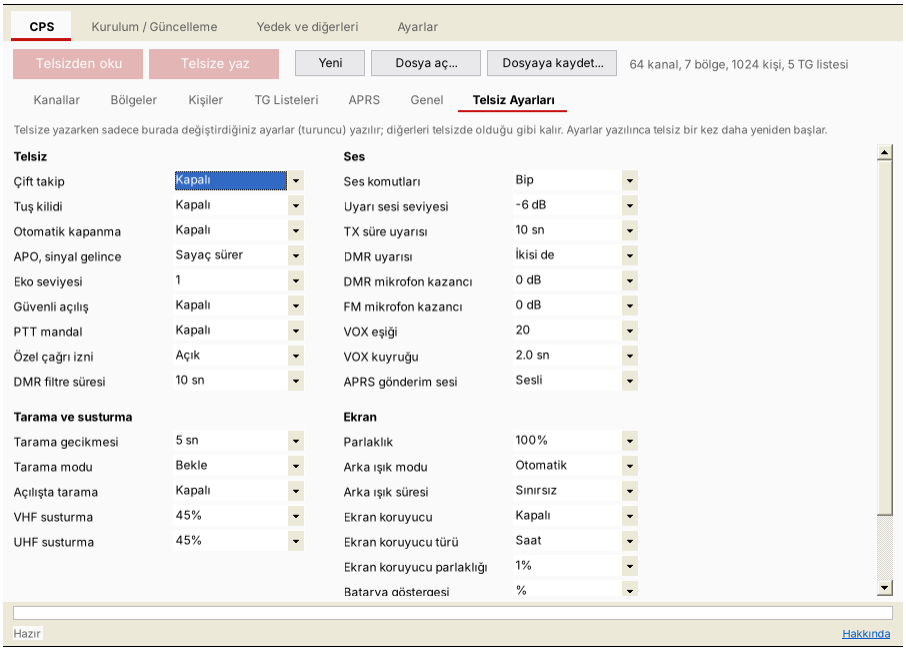


CPS › APRS (Turkish interface)

- There is **one APRS config**: name, sender SSID, path (WIDE1-1, WIDE2-1), **beacon frequency**, symbol, comment, fixed position, position ambiguity (sends the position rounded by about 200 m – 100 km).

- When APRS is on, the position is sent on the **beacon frequency** whatever the open channel. Without an APRS config, a **Create the APRS config** button is shown; 144.800 MHz is preset.
- The **Symbol** button opens the APRS symbols as pictures (primary table / and alternate table \); click the one you want.
- APRS on / off, the interval and the APRS sound are set in the radio menus (or in CPS › Radio Settings).

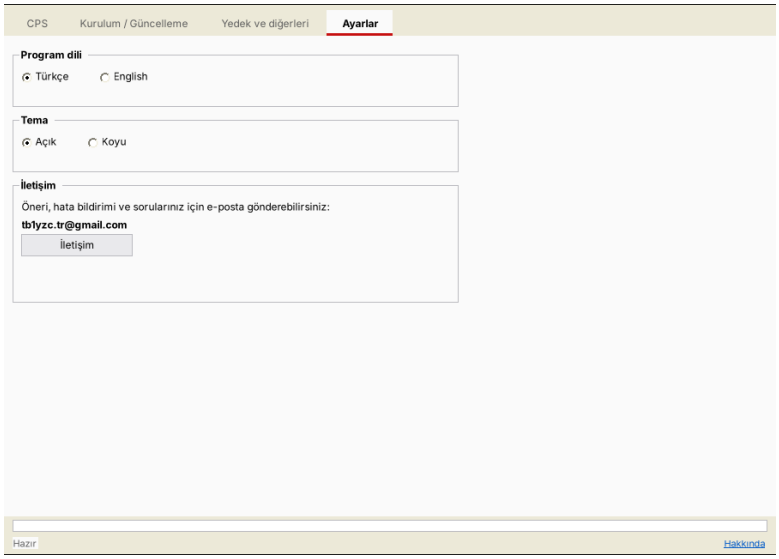
Radio Settings



CPS › Radio Settings (Turkish interface)

- Most of the radio menu settings can be set here: dual watch, keypad lock, auto power off, eco, PTT latch, private calls, DMR filter time, scan and squelch, voice prompts, beep volume, mic gains, VOX, APRS sound, brightness, backlight, screen saver, battery display, LEDs, time zone.
- **Only the settings changed in the CPS** (their name turns orange) are written; the others stay as they are in the radio. An old file or a new empty codeplug does not overwrite the settings made in the radio menus.
- When the settings are written the radio restarts twice (codeplug, then settings); the program reconnects by itself.

7. Program settings



- **Program language:** Türkçe or English.
- **Theme** (Windows): Light or Dark. The Android app always uses the dark theme.
- **Contact:** the Contact button opens your e-mail program for suggestions, bug reports and questions. Issues can also be reported on `github.com/TB1YZC-Lab/TA390`.

8. Keys and controls

Control	Description
Power / volume knob	Switches the radio on and off, sets the volume.
Top channel knob	Channel screen: channel (zone with Dual Watch on); VFO: frequency step. Changes values in the menus.
PTT	Push to talk.
SK1 (above PTT)	Channel info and SK1 shortcuts. SK1 + Green: Quick Menu.
SK2 (above SK1)	Modifier key. Long press: monitor (opens the squelch).
Green	Menu / confirm / save.
Red	Back / cancel. On the channel screen: goes to the VFO.
Up / Down	Menu navigation. Channel screen: TG (DMR) or squelch (FM).
0–9, *, #	Digit and letter entry, shortcuts.

In this manual, "**SK2 + #**" means: hold SK2 and press the other key. A long press is about 0.5 s, a "very long" press about 1.2 s. All the shortcuts are also listed on the radio in Main menu › **Shortcuts**.

9. Display

Header line



DMR channel: mode, TS1 and colour code (C1), power, APRS icon, GPS icon, battery



FM channel: tones (CTR), Talkaround (RA), power, APRS icon, GPS (no fix), battery

Item	Meaning
DMR / FM	Mode of the channel. Struck through: the channel is muted. Framed: DMR filter on.
N	FM narrow band (12.5 kHz), shown to the right of FM; not shown for wide band.
TS1 / TS2, C1	DMR timeslot and colour code. Framed: TS / CC filter on; bold TS: timeslot changed with *. On FM this place shows the tones: CT / DT (TX tone CTCSS / DCS), C / D (RX tone only), R (RX tone too).
Power	Transmit power.
((o))	APRS icon: shown only when APRS beaconing is on, red while a position is sent.
Location mark	GPS icon: shown when the GPS is on. Filled with a fix, empty without.
RA	Talkaround on (on a repeater channel, direct talk on the RX frequency). Right of the tones on FM, in place of TS on DMR.

Item	Meaning
85% / 7.9V	Battery (Options › Display › Battery: percent or volts). Bold when auto power off is on.

Under the header: signal strength bar while receiving, microphone level bar while transmitting.

Middle and bottom

Middle: channel name or VFO frequency. On a DMR channel, like the Dual Watch rows, the **TG list / talkgroup** is shown in small text under the channel name; on a DMR call the callsign / name of the caller is shown. Bottom: zone name (on FM with the channel number). On the VFO only the RX frequency is shown (TX frequency while transmitting). Turning the volume knob shows a volume bar for a moment (Display › Volume).

Screen wake up and screen saver

- When the screen is dark, the first key press, SK1 / SK2 or knob step **only turns the screen on**; the channel does not change by mistake. PTT is not affected.
- With **Scr. saver** on, the screen does not go fully dark when the backlight goes off; it stays at the **Svr bright** level (default 1%). **Svr. mode**: **Screen** shows the normal screen dimmed; **Clock** shows a big clock with the date, day and battery percent below. A red phone icon next to the battery shows a missed call. The normal screen comes back on a signal.
- For the screen saver the backlight mode must be Auto, Squelch or Buttons, and the Timeout not 0. Without the clock set, lines are shown instead of digits.
- After saving settings “Saved” is shown for 1 second; any key closes it at once.

10. Channel screen

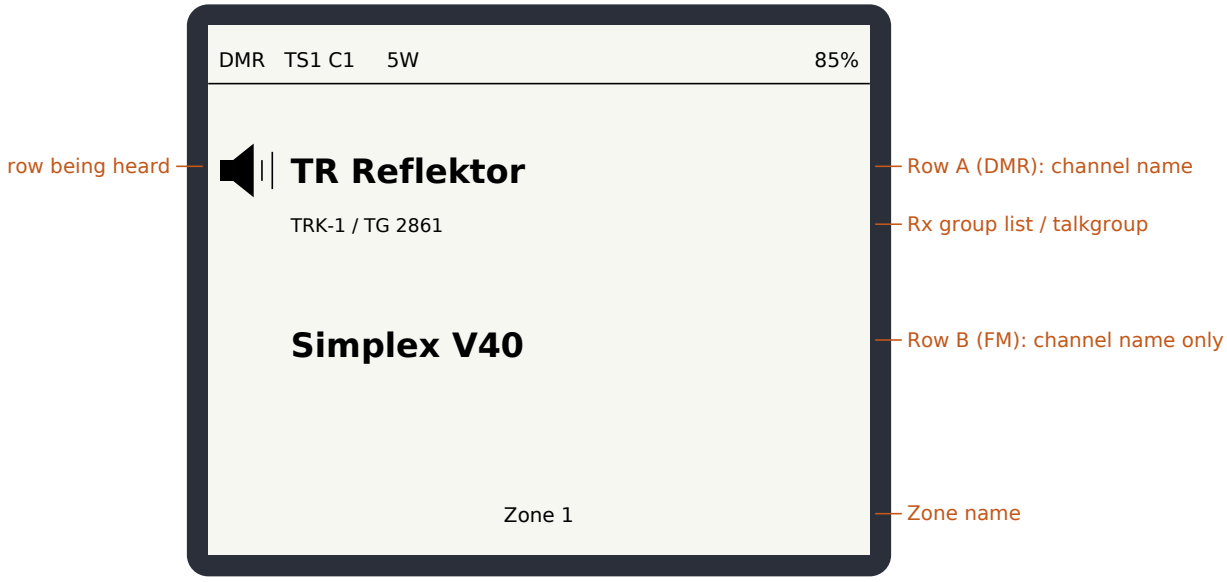
Main screen, working with the channels of the codeplug. The table is for **Dual Watch off**; the Dual Watch keys are in section 11.

Key	Function
Knob	Next / previous channel (in the zone).
SK2 + knob	Next / previous zone.
Up / Down	DMR: talkgroup of the TG list. FM: squelch level.
SK1 + Up / Down	DMR: changes the TG list of the channel and saves it.
SK2 + Up / Down	Transmit power.
0–9, Green	Goes to a channel number. **Red** cancels the entry.
Green	Main menu.
SK2 + Green	Channel details (section 20).
SK1 + Green	Channel Quick Menu (Dual Watch, Talkaround, power...).
Red	VFO screen.
SK1	Channel info (frequency, tone, colour code...). **Red** closes it.
SK1 very long	With voice prompts on, reads the channel info.
SK2 long	Monitor: squelch open while held.
SK2 (DMR call on screen)	Sets the TX TG, timeslot and colour code to the received call (to answer).
*	DMR: TS1 ↔ TS2. FM: bandwidth narrow ↔ wide.
* **long**	DMR: cancels the timeslot change.
SK2 + *	Temporarily switches the channel DMR ↔ FM.
#	TG / private call (PC) / contact entry.
SK2 + #	Contact quick list.
# long	Reverse repeater on / off (RX and TX swapped).
8 long	Bearing and distance to the location stored in the channel.
SK1 + 1	APRS on / off.
SK1 + 2	GPS on / off.
SK1 + 3	Mutes / unmutes this channel only; the other channels are heard. All channels are audible after power on.
SK2 + 1	Sends the APRS position at once (APRS must be on).
SK2 + 2	Zone scan: scans the channels of the zone, stops on a channel with a call. **Red** cancels.
Green, then *	Keypad lock on / off (pressing both together works too).
SK1 + Red	Day / night theme.
SK2 + Red	Leaves the private call, back to the previous TG.

11. Dual Watch

Two stored channels of the current zone are shown one under the other; the receiver switches between them all the time and monitors both. The **arrow** shows the selected channel: PTT transmits on it. The audio comes from the channel with a signal.

On / off: SK1 + Green › **Dual Watch** row, knob: On / Off, **Green** to confirm. It can also be set in CPS › Radio Settings. The two channels and the arrow come back after power off / on.



Example: row A is selected and is the one being heard (speaker instead of arrow).

- Under a DMR channel name the TG list / talkgroup is shown; an FM channel shows its name only. The zone name is at the bottom.
- A **speaker** icon appears on the row with a signal (in place of the arrow on the arrow row).
- **Missed call:** a red phone icon right of the row that had a call; cleared by any key on the channel screen.

Key (Dual Watch on)	Function
Knob	Next / previous zone.
Down / **Up**	Next / previous channel on the arrow row.
SK2 short	Moves the arrow to the other row.
SK2 2 s	Monitor (arrow channel).
SK2 + Up / Down	Talkgroup of the arrow row (DMR).
SK1 + Up / Down	TG list of the arrow row (DMR).
SK1	Info of the arrow channel.
PTT	Transmits on the arrow channel.

- A key press stops the monitoring; it goes on 3 seconds after the last key. After a signal or a transmission the receiver stays on that channel for the **Scan delay** time, so an answer is not missed.
- With a DMR and an FM channel on the same frequency, the audio always comes from the arrow row.

- With Dual Watch on, the power is changed in Quick Menu › Pwr. Dual Watch works with stored channels only.

12. VFO screen

For frequencies entered by hand and for scanning. There is a single VFO. **Red** on the channel screen goes to it, **Red** goes back (while scanning the first **Red** stops the scan, the second goes back to the channel screen). Only the **RX frequency** is shown and the radio **listens on it**; no other channel is monitored. **Transmitting** uses the **TX frequency** set in the VFO details (**SK2 + Green**): at first it is the same as the RX frequency; when the RX frequency changes the offset is kept. From top to bottom: frequency, signal strength (S / dBm), step (e.g. Step: 12.5 kHz, with the TX offset, e.g. TX -7.600; TX 0 when RX and TX are the same), scan line and stored channel name. The VFO details have no name row. No TG on the VFO: in DMR mode only the presence of DMR traffic is checked. No TG on the screen, TS and CC are shown only while transmitting; the data of a received call are always shown. **All the filters are cleared each time the VFO is entered** (FM tone, DMR TG, CC and TS filters), so the scan catches every signal. They can be set by hand in the VFO Quick Menu; leaving and entering the VFO clears them again. The channel filters are not affected. **Signal strength:** shown above the frequency (e.g. S7 -101 dBm, S9+20 -73 dBm). **Stored channel name:** if the frequency is the same as a channel of the codeplug, its name is shown at the bottom (also when a scan stops).

Key	Function
0–9	Frequency entry. **Green** applies early, **Red** cancels.
Knob	One step. **SK1 + knob**: 1 MHz.
7 long	Changes the frequency step (the Step line changes).
8 long	Next TX shift value downwards (like the knob in the channel details: 0, 0.6, 1.0, 1.5, 1.6, 2.0, 4.6, 5.0, 7.0, 7.6, 9.0, 9.4 MHz).
9 long	Next TX shift value upwards (same values).
6 long	**Save channel:** after a confirmation, the frequency (FM or DMR, with tone / colour code / timeslot / TG) is saved as a new channel in the current zone; its name is the frequency (e.g. 145.5000), changeable in the channel details.
Up / Down	Without a scan: squelch on FM, talkgroup on DMR.
SK2 + Up / Down	Transmit power.
1 long	**Step scan:** from the current frequency with the set step; at the end of the band it goes on from the start of the band.
2 long	**Range scan:** enter the lower and upper frequency (6 digits each); **Start** is shown above the **Green** key. **Green** starts, **Red** cancels.
3 long	**FM / DMR scan** switch (also while scanning). The DMR scan is as fast as on FM and catches DMR traffic.
4 long	FM narrow (12.5 kHz) ↔ wide (25 kHz). “N” in the header for narrow.
5 long	**Filter scan** (for the displayed frequency). FM: finds the CTCSS / DCS tone of the other station and puts it into the VFO. DMR: waits for traffic (“CC/TS search”), puts its **colour code, timeslot and TG** into the VFO. Any key cancels.
Up / Down (scanning)	Scan direction: up / down.
***** **(scanning, stopped)**	Skips this frequency for this scan.

Key	Function
Red (scanning)	Stops the scan; pressed again: channel screen.
# long	Spectrum (section 13).
*	DMR: TS1 ↔ TS2. Nothing on FM (bandwidth: **4** long).
#, SK2 + *	Like on the channel screen.
SK2 + Green	VFO details (tone, step...).
SK1 + Green	VFO Quick Menu.
SK1 + SK2 held	Listens on the repeater input.
SK1 + 1 / SK1 + 2 / SK1 + 3	APRS, GPS on / off; mute / unmute the VFO.
SK2 + 1	Sends the APRS position at once.
Green, then *	Keypad lock on / off.

13. Spectrum

Opened with a long **#** on the VFO screen; shows the signals of a frequency range as a graph.

Key	Function
Knob	Moves the centre frequency.
SK2 + knob	Span (range).
Up / Down	Gain.
SK2 + Up / Down	Noise floor.
SK1 + Up / Down	Resets gain and floor.
4 / 6	Moves the cursor. **5**: cursor to the strongest signal.
Green	Tunes to the cursor frequency and listens. While listening **Green** stays on it, **Red** skips the signal.
0	Waterfall on / off.
8	Stop on signal on / off (goes on 3 s after the signal).
*	Bandwidth (FM). **SK2** long: monitor. Other keys leave the spectrum.

14. Using DMR

- Each DMR channel has a **TG list** or a single contact. The talkgroup is selected with **Up / Down**, the TG list is changed with **SK1 + Up / Down**. The talkgroup can also be chosen in the **TG** row of the channel details (section 20).
- **#** opens the entry screen; **#** again cycles TG → PC (private call) → Contact. Type the number and apply with **Green**. A TG entered by hand is cancelled when the channel changes.
- **Private call:** with Options › Radio › **Allow PC** on, “Return call to” is asked; **Green** accepts, **Red** refuses. **SK2 + Red** goes back to the previous TG.

- **DMR Filter** (Quick Menu): which calls are heard: off, the selected TG only, contacts or the TG list.
- **TS Filter**: when off, calls on the other timeslot are heard too. **CC Filter**: colour code filter; when off, every colour code is heard.
- **Filter time** (Options › Radio): how long the filter stays locked to a call after it.
- Received DMR calls are stored in Main menu › **Logs** › **DMR** (section 24).

15. Using FM

- **Squelch: Up / Down**. The levels are per band in Options › Radio (no per channel squelch); a change with Up / Down is reset when the channel changes.
- **Bandwidth: *** switches **narrow** (12.5 kHz) ↔ **wide** (25 kHz).
- **CTCSS / DCS:** in the channel details or in the CPS. On the VFO a **5 long** press finds the tone of the other station.
- **1750 Hz: SK2** while transmitting on FM (repeater access). **DTMF:** while transmitting the digit keys send DTMF tones.
- **Talkaround** (Quick Menu): direct talk on the RX frequency on a repeater channel. **RA** is shown in the header when on.
- Received FM calls (squelch open with audio) are stored in Main menu › **Logs** › **FM**.

16. Scanning

- **Zone scan:** on the channel screen **SK2 + 2** scans the channels of the zone; it stops on a channel with a call and goes on when the call ends. **Red** cancels. **Up** / knob resume, **Down** reverses the direction, while stopped * skips that channel for this scan. Other keys stop the scan; the first PTT only stops it.
- **Zone Skip / All Skip** in the channel details / CPS remove a channel from the scan permanently.
- Options › Radio: **Scan delay** (wait after a signal; each channel is checked for a fixed 30 ms), **Scan mode** (what to do on a signal), **Scan On Boot**.
- **VFO:** step scan (**1** long), range scan (**2** long, **Green** starts), FM / DMR scan (**3** long), spectrum (**#** long). See section 12.

17. Quick menus

Opened with **SK1 + Green**. **Up / Down** selects the row, the knob changes the value, **Green** applies, **Red** cancels.

The Quick Menu settings apply to the selected channel only (Filter, DMR Filter, CC / TS Filter, Talkaround, Mute, Power). The other channels use the general settings; back on the channel, its setting applies again. In Dual Watch each row uses its own settings. These settings are not saved: after power off / on all channels start with the general settings (Options menu). Dual Watch on / off is general.

Channel Quick Menu	Function
VFO --> Channel	Writes the VFO settings into the channel (asks for a confirmation).
Filter	FM tone filter (this channel).
DMR Filter	DMR call filter (off, TG, contacts, TG list), this channel.
CC Filter	Colour code filter (this channel).
TS Filter	Timeslot filter (this channel).
Talkaround	Direct talk on the RX frequency (repeater channels only, this channel).
Mute	Mutes this channel only; its name is struck through. Other channels and Dual Watch rows are not affected.
Dual Watch	Dual Watch on / off (general).
Pwr	Transmit power of this channel (also over the channel's own power).

VFO Quick Menu	Function
Filter, DMR Filter, CC Filter, TS Filter	Like the Channel Quick Menu; for the VFO only.
Mute	Mutes the VFO only.

18. Quick keys

- On a menu row, **SK2 + digit** assigns the row to a quick key. A zone in the zone list or a contact in the contact list can be assigned too.
- On the channel or VFO screen, **SK2 + digit** (short) runs the quick key; a long press deletes it.
- **SK2 + 1** and **SK2 + 2** are not quick keys: they send the APRS position at once / start the zone scan.

19. Main menu

Opened with **Green**. **Red** goes back, a long **Red** goes back to the channel / VFO screen. In the main menu * locks the keypad, # locks / unlocks the PTT.

Menu	Content
Zone	Zone selection (All Channels is listed too).
Contacts	All the DMR contacts in one list (group, private, all call), with + New contact at the top.
TG lists	Adds / removes talkgroups of the TG lists (section 21).
Channel details	All the settings of the current channel (section 20).
APRS message	Write an APRS short message, sent messages (section 22).
Gps/Location	GPS on / off, position, number of satellites, heading (section 23).
Options	General, Radio, Display, Sound, APRS, Language, Calibration (section 27).
Logs	DMR and FM call logs (section 24).
Games	Snake, Tetris, Mines (section 25).
Satellite track	Amateur satellite passes and tracking (section 23).
Shortcuts	List of all the key shortcuts.
Radio info	Battery, temperature, clock, date, location, uptime, alarm (section 26).
Firmware info	TA390 name, version and build date.

20. Channel details

SK2 + Green or Main menu › Channel details. **Up / Down** selects the row, the knob changes the value. Digits are typed into the frequency, DMR ID and location fields. **Green saves the channel** (“Saved” is shown, the screen stays open); **Red** leaves.

Setting	Description
Name, Rx / Tx, Mode	Channel name, frequencies, DMR or FM.
Location, Lat, Lon	Location of the channel (for bearing / distance). In Lat / Lon a long * puts the GPS position.
DMR ID	DMR ID of this channel; the radio ID is shown when the channel has none.
Color Code, Timeslot	DMR colour code (0–15) and TS1 / TS2.
TG Lst, Contact	TG list of the channel, or (without a TG list) its single contact. Selecting one sets the other to None.
TG	On a DMR channel, the talkgroup of the TG list. The knob browses the TGs of the list; saved with **Green**, the channel opens with this TG (the same as choosing it with **Up / Down** on the channel screen).
Rx / Tx tone	CTCSS / DCS.
BW, Step	Narrow (12.5 kHz) / Wide (25 kHz). Step only in the VFO details (VFO frequency step).
TOT	Transmit time limit.

Setting	Description
Rx Only	Blocks transmitting on the channel.
Zone Skip, All Skip	Skipped by the scans.
VOX	Voice operated transmit on the channel.
Ch Power	Transmit power of the channel. Without a channel power, the Quick Menu › Pwr value is shown and used.
Force DMO	Forces direct mode on DMR.

21. TG lists

- Main menu › **TG lists**: talkgroups are added to / removed from the TG lists of the codeplug without a computer.
- In the list **Green** opens a list. On a TG **Green** asks “Remove?”. The “+ Add” row lists the TG contacts of the codeplug (the ones already in the list are marked with *), **Green** adds.
- 32 TGs per list maximum, at least one TG must stay. Changes are saved at once; writing an old codeplug from the CPS overwrites them.

22. APRS and APRS message

- APRS beaconing is **general**: when Options › APRS › **Mode** is not Off (Manual, PTT, Auto, Smart), the position is sent **whatever the open channel** on the frequency of the **first APRS config with a beacon frequency** of the codeplug (CPS › APRS); then the radio goes back to its channel.
- The **((o))** icon in the header is shown when beaconing is on, red while a packet is sent.
- With the GPS as position source, nothing is sent before a GPS fix. **Location** setting: Channel (fixed position) or GPS.
- **Interval**: first beacon interval. **Decay**: longer interval when not moving. **Compress**: compressed position. **Smart** mode depends on speed and turns: Slow / Fast Rate, Low / Hi Speed, T. Angle / T. Slope / T. Time.
- **Power**: APRS transmit power. **Audio**: Audible: the packet is heard on the speaker, Silent: it is not. **DMR TA**: data sent with the voice during DMR calls, on all channels: Off, Location (GPS), Text or Both.
- **SK2 + 1** sends the position at once (in every mode; the only way in Manual mode). **SK1 + 1** turns beaconing on / off. No beacon during a zone scan; beacons are sent with Dual Watch. The keys do not react for a moment while a packet is sent.

APRS message

Main menu › **APRS message**: **Write message** and **Sent messages**. A message is sent once, with the same APRS config and frequency as the position packets. The radio cannot receive APRS, so answers and acknowledgements are not shown.

Key (Write message)	Function
Up / Down	Switches between the To and Message fields.
0–9, *	Letter entry (press the same key repeatedly). **0**: space first, then 0. To: capitals A–Z, 0–9, -.
Knob	Moves the cursor. **SK2 + knob (−)** deletes a letter.
#	Deletes the letter left of the cursor; long press clears the field.
Green	Sends (“Sent” / “Not sent”).
Red	Back; the text is cleared.

- To: 9 characters maximum (TA1ABC-9), message: 67 characters maximum. Turkish letters and | ~ { cannot be used.
- **Sent messages:** last 15 messages, newest first. **Green** opens the details (to, date, time, message); in the details **Green** deletes the message (with a confirmation). The list is kept after power off.

23. Gps/Location and Satellite track

- Main menu › **Gps/Location**. With the GPS off, a green **Gps On** is shown above the Green key.
- **Green** turns the GPS on / off (**SK1 + 2** does the same on the channel and VFO screens). Knob right / left: **Off → On → NMEA** (NMEA: the position data are also sent to the computer over USB).
- **Up / Down** changes the page: coordinates and accuracy, satellites, altitude, heading / speed.
- With the GPS on, the header shows a location icon: filled with a fix, empty without. The first fix can take a few minutes outdoors.
- **Satellite track:** amateur satellite pass predictions and tracking. Location and clock must be set; otherwise “Gps/Location waiting” is shown. The satellite data are uploaded with the TA390 program (section 5). Options › General › **Satellite**: Auto selects the next satellite by itself, Manual lets you choose.

24. Logs

- Main menu › **Logs** › **DMR** or **FM**. Each received call is stored when it ends; newest first.
- **DMR log:** caller (name / ID), TG or PC, timeslot, frequency, channel, date, time, length.
- **FM log:** frequency, channel, tone (CTCSS / DCS), date, time, length.
- **Up / Down** or the knob scroll, **Green** opens the details, a long **Green** deletes all the logs (with a confirmation). In the details **Green** deletes that log (with a confirmation), **Red** goes back to the list.
- Each list keeps 512 logs; when full the oldest are deleted. The logs are kept after power off. With a very large DMR ID database there is no room for the logs and this is shown.

25. Games and Shortcuts

Main menu › **Games**. The radio keeps listening while playing. The best scores (Best) are kept.

Game	Keys
Snake	**2** / **Up**, **8** / **Down**, **4**, **6**: direction.
Tetris	**4** / **6** or the knob: left / right. **2** / **Up**: rotate. **8** / **Down**: one row down. **5**: drop.
Mines	**2 4 6 8**, Up / Down or the knob: cursor. **5** / **Green**: open. * / #: flag.
All	**Green**: pause / resume (not in Mines). **Red**: back to the game list. After game over **Green** restarts.

Main menu › **Shortcuts**: all the key shortcuts are listed on the radio screen under Channel, VFO, Spectrum and General (in English when the radio language is English).

26. Radio info

Main menu › **Radio info**: battery level and graph, temperature, clock, date, location, uptime and alarm time. Clock, date and location can be entered here by hand; the clock can also be set from the TA390 program.

27. Options

Main menu › Options. **Up / Down** selects the row, the knob changes the value. **Green** saves and stays in the menu, **Red** cancels the changes and leaves. Most settings can also be set in CPS › Radio Settings.

General

Setting	Description
Auto lock	Locks the keypad after a time without key presses (Off or minutes).
Hotspot	MMDVM hotspot mode (use with a computer).
Eco Level	Power saving. A higher level saves battery but may detect signals later. Off while Dual Watch is running.
Safe Pwr-On	When on, the radio only powers on with SK1 held.
APO, APO with RF	Auto power off time; with APO with RF a received signal restarts the timer.
Satellite	Auto / Manual for satellite tracking.

Radio

Setting	Description
Band Limits	Transmit frequency limits.
TX Inhibit	Blocks all transmissions.
Filter time	How long the radio stays locked to a DMR call.
Scan delay	Wait after a signal while scanning (used by Dual Watch too).
Scan mode	What to do on a signal.
Scan On Boot	Scan at power on.
VHF / 220 / UHF Squelch	Default FM squelch per band.
PTT latch	PTT press on / press off mode.
Allow PC	Received private calls: Off, On, PTT.
User Power	User power for the highest power step.
DMR crc	DMR data check.
Mode	Power version of the radio: 5 W or 10 W.

Display

Setting	Description
Text size	Normal or large text.
Brightness	Screen brightness.
Svr bright	Screen saver brightness (1% – Brightness).
Mode	Backlight: Auto, Squelch, Manual, Buttons, None.

Setting	Description
Timeout	Backlight off time.
Scr. saver, Svr. mode	Screen saver on / off; Screen or Clock.
Theme	Day or Night.
Order, Contact	Source of the caller name on DMR; 1 or 2 lines.
Battery	Battery display: % or volts.
Volume	Bar shown when the volume knob is turned.
LEDs	Transmit / receive LEDs.
Timezone, UTC	Time zone; clock shown in local time or UTC.

Sound

Setting	Description
Prompt	Voice menu prompts: Silent, Beep, voice levels.
Timeout beep	Warning beep before the TOT.
Beep vol	Level of the key and warning beeps.
DMR beep, RX beep, Talker	Beeps at the start / end of transmission, at the end of reception and when a talker starts.
DMR mic, FM mic	Microphone gain.
DMR Rx AGC	Automatic level of the DMR received audio.
VOX Thres., VOX Tail	Voice transmit threshold and release delay.

APRS, Language, Calibration

- **APRS**: section 22.
- **Language**: Turkish or English (the language flashed with the TA390 program).
- **Calibration**: calibration pages: power, VHF / UHF frequency, **temperature and battery calibration** and factory reset. **Green** saves. Do not change anything unless you know what you are doing; make a calibration backup first.

28. Locks, power off, power on

- **Keypad lock:** while locked, a lock icon is shown in the header next to the APRS icon; a key press shows “locked” for 2 seconds. **Green** then * locks; while locked **Green** then * within 2 seconds unlocks (pressing both together works too). Automatic lock: Options › General › Auto lock.
- **PTT lock:** # in the main menu.
- **Power off:** when the power knob is turned off, a message is shown for a moment; turning it back on in that time cancels the power off.
- **Power on:** an uploaded boot image is shown full screen; otherwise the TA390 boot screen. A key skips it.

Hold while switching on	Result
SK1	With Safe Pwr-On on, the radio only powers on this way.
SK2	Resets the radio settings to the defaults.
SK2 + Down	Deletes the uploaded boot image and melody.
PTT + SK1	Bootloader mode (firmware flashing).

29. Troubleshooting

Symptom	Solution
The program does not find the radio (switched on normally)	Check the USB cable and the COM port; close the OpenGD77 CPS or other programs using the port. Android: check the USB permission and Settings › USB check.
No radio found in bootloader mode	Switched on with PTT + SK1? Is the STM DFU driver installed (OpenGD77 CPS)? For an STDFU.dll error see section 3.
A key does nothing	With a dark screen the first press only turns the screen on. While an APRS packet is sent the keys do not react for a moment. Check the keypad lock.
PTT does not work	PTT lock, TX Inhibit, Rx Only, band limits or scanning (the first PTT stops the scan).
No DMR audio	Flashed with the donor? Colour code, TS, TS Filter, DMR Filter settings.
APRS is not sent	APRS Mode on? An APRS config with a beacon frequency in the CPS? GPS fix?
Satellite track does not open	Location and clock must be set; upload the satellite data with the TA390 program.
Settings did not change after writing from the CPS	Only the radio settings changed in the CPS (orange) are written.
Going back to an old firmware	Restore the MCU ROM backup or flash the old .bin file.

30. Version and licence

v1.3 changes

Channel screen

- Mute is per channel (SK1+3 or Quick Menu): the name of the muted channel is struck through, the other channels are heard. All channels are audible after power on.
- The Quick Menu settings (FM / DMR filters, CC / TS filter, Talkaround, Mute, Power) apply to the selected channel only; they are not saved.
- The single channel DMR screen looks like Dual Watch: channel name, TG list / TG in small text below it, only the zone name at the bottom.
- SK2+2: zone scan (stops on a channel with a call, Red cancels); the Up long press scan was removed.

- "Channel --> VFO" and the 0 long press mute shortcut were removed.

VFO screen

- Single VFO (A / B removed): it listens on the displayed frequency and transmits on the TX frequency of the VFO details (automatic repeater shift and SK1+0 removed). Centred frequency, no "R" letter, no name row in the VFO details.
- 1 long: step scan (whole band), 2 long: range scan (Green = Start), 3 long: FM / DMR scan, 4 long: narrow / wide, 5 long: filter scan (FM tone, DMR CC / TS / TG), 6 long: save as a channel.
- While scanning Up / Down sets the direction, Red stops (pressed again: channel screen). The DMR scan is as fast as the FM scan and stops on DMR traffic.
- All the filters are cleared each time the VFO is entered.
- VFO screen: frequency, signal strength (S7 -101 dBm), step and TX offset (TX -7.600, TX 0 when the same), 7 long: change the step (SK1+5 removed), 8 long / 9 long: TX shift - / + (the knob values of the channel details: 0.6 ... 9.4 MHz), scan line, and the channel name when the frequency belongs to a stored channel.
- Removed from the VFO Quick Menu: Tx/Rx rows, VFO A/B, Dual Watch, Tone scan, Freq. bind and VFO --> New channel. No TG in the VFO: in DMR mode only DMR traffic is searched; no TG on the screen. SK1+4 and the Up long scan removed. * does not change the bandwidth in FM.

Header line

- FM narrow band is shown as a separate "FM N".

Shortcuts and other

- Keypad lock: a lock icon in the header while locked, "Keys locked" for 2 seconds. Green then * locks; while locked Green then * unlocks. SK2+* unlock removed.
- SK2+1: sends the APRS position at once; the messages are on two lines to fit the screen.
- APRS packets use the device code APTA39.
- Sound options: DMR Rx AGC moved below FM mic.
- The shortcut list of the radio (Turkish and English) was updated.
- Channel details: new TG row: on a DMR channel the TGs of the TG list are browsed with the knob; saved with Green, the channel opens with that TG (no shortcut needed).
- Channel details: Step (kept in the VFO details), Squelch, Beep, Eco and TA Tx TS1 / TS2 removed: squelch, beeps and Eco come from the general settings; the DMR ID row shows the radio ID when there is no channel ID.
- Main menu > Contacts opens all the DMR contacts directly (group, private, all call in one list), with + New contact at the top; FM DTMF contacts removed from the menu.
- The frequency details shown with SK1 use Rx / Tx instead of R / T.
- APRS options: DMR TA added (all channels): Location (GPS), Text or Both sent during DMR calls.
- Radio options: Scan dwell removed: scan and Dual Watch look at each channel for a fixed 30 ms (fastest). Power saving (Eco) is fully off while Dual Watch is running.

Main menu › **Firmware info** shows “TA390 MD-UV390”, the version and the build date. The version in the file name (`TA390_v1.3.bin`) is the same as on the screen. The Windows program and the Android app always have the same version number as the firmware.

New versions and issue reports: `github.com/TB1YZC-Lab/TA390` (Releases and Issues). Do not contact the OpenGD77 team or forum about TA390 problems.

OpenGD77 is the open source project of Roger Clark VK3KYY, Daniel Caujolle-Bert F1RMB and the contributors. TA390 is derived from it and is not an official OpenGD77 release. It is free of charge and may not be used for commercial purposes. The firmware and the programs are provided as is, without any warranty; use them at your own risk.

The licence texts are in the `LISANSLAR` folder of the packages: OpenGD77, FreeRTOS (MIT), ST USB / HAL, CMSIS (Apache 2.0), libusb (LGPL-2.1) and the APRS symbols (aprs.fi symbol set, OH7LZB; brand logos removed).

By TB1YZC · `github.com/TB1YZC-Lab/TA390`